MODULE4

Python Data & APIs

• NumPy arrays and calculations

• Pandas and DataFrames

• Data cleaning and manipulation

• Matplotlib and Seaborn charts

• REST APIs, HTTP requests, and JSON

• Python Data Analysis Application mini project

NumPy

What is NumPy?

NumPy means Numerical Python. It is a Python library used for numerical calculations, arrays, and mathematical operations.

Why Use NumPy?

• Work with numerical data

• Perform calculations on many values

• Find totals, averages, minimums, and maximums

Creating an Array

```
Code:
import numpy as np
marks = np.array([80, 75, 90, 85, 95])
print(marks)
```

Output:

[80 75 90 85 95]

Accessing Array Elements

```
Code:
print(marks[0])
print(marks[2])
```

Output:

80

90

Array indexing starts at 0.

Array Operations

```
Code:
numbers = np.array([10, 20, 30, 40])
print(numbers + 5)
print(numbers * 2)
print(numbers - 5)
```

Output:
[15 25 35 45]
[20 40 60 80]
[5 15 25 35]

Useful NumPy Functions

```
Code:
marks = np.array([80, 75, 90, 85, 95])
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
```

Output:
Total: 425
Average: 85.0
Highest: 95
Lowest: 75

Pandas and DataFrames

What is Pandas?

Pandas is a Python library used for data analysis, data cleaning, data manipulation, and working with tables and CSV files.

What is a DataFrame?

A DataFrame is a table containing rows and columns.

Creating a DataFrame

```
Code:
import pandas as pd
```

```
data = {
        "Name": ["Pujitha", "Anu", "Ravi"],
        "Age": [19, 20, 19],
        "Marks": [85, 90, 75]
}
df = pd.DataFrame(data)
print(df)

Output:
        Name  Age  Marks
0  Pujitha   19     85
1       Anu   20     90
2       Ravi   19     75
```

Data Cleaning and Visualization

What is Data Cleaning?

Data cleaning means finding and fixing missing, duplicate, or incorrect data.

Checking Missing Values

```
Code:
print(df.isnull())
print(df.isnull().sum())
```

Removing or Filling Missing Values

```
Code:
df = df.dropna()  # remove rows with missing values
df["Marks"] = df["Marks"].fillna(0)  # fill missing marks with 0
```

Choose the approach that makes sense for the data.

Removing Duplicates and Sorting

```
Code:
df = df.drop_duplicates()
df = df.sort_values("Marks")
df = df.sort_values("Marks", ascending=False)
```

What is Matplotlib?

Matplotlib is a Python library used to create graphs and charts.

Line Chart

Code:

```
import matplotlib.pyplot as plt

names = ["Pujitha", "Anu", "Ravi"]
marks = [85, 90, 75]

plt.plot(names, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

Output: A line chart showing each student's marks.

Bar Chart

Code:

```
plt.bar(names, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

Output: A bar chart comparing student marks.

Pie Chart

Code:

```
plt.pie(marks, labels=names, autopct="%1.1f%%")
plt.title("Student Marks")
plt.show()
```

Output: A pie chart showing each value as a slice.

What is Seaborn?

Seaborn is a Python visualization library built on Matplotlib. It helps create attractive statistical graphs.

Seaborn Bar Plot

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(x="Name", y="Marks", data=df)
plt.title("Student Marks")
plt.show()
```

REST APIs and JSON

What is an API?

API means Application Programming Interface. It allows one application to request data or services from another application.

What is a REST API?

A REST API is a common way for applications to communicate over a network using HTTP methods.

Common HTTP Methods

GET — retrieve data

POST — send or create data

PUT — update or replace data

DELETE — delete data

What is JSON?

JSON means JavaScript Object Notation. It is a common format used to exchange data between applications.

Make an API Request

Code:

```
import requests

url = "https://jsonplaceholder.typicode.com/users"
response = requests.get(url, timeout=15)
print(response.status_code)
```

Output:

```
200
```

A status code of 200 means the request succeeded. Internet access is required.

Mini Project

Project Title

Student Data Analysis Application

Python Code

```
import pandas as pd
 import matplotlib.pyplot as plt

# 1. Create dataset
 data = {
        "Name": ["Pujitha", "Anu", "Ravi", "Kiran", "Sita"],
        "Age": [19, 20, 19, 21, 20],
        "Marks": [85, 90, 75, 88, 95]
 }
 df = pd.DataFrame(data)

 # 2. Clean data
 df = df.drop_duplicates()
 df["Marks"] = pd.to_numeric(df["Marks"], errors="coerce")
 df = df.dropna(subset=["Marks"])

 # 3. Analyze data
 average = df["Marks"].mean()
 highest = df["Marks"].max()
 lowest = df["Marks"].min()
 top_student = df.loc[df["Marks"].idxmax()]

 print("STUDENT DATA")
 print(df)
 print("\\nAverage Marks:", round(average, 2))
 print("Highest Marks:", highest)
 print("Lowest Marks:", lowest)
 print("\\nTop Student:")
 print(top_student)

 # 4. Create and save chart
 plt.bar(df["Name"], df["Marks"])
 plt.xlabel("Students")
 plt.ylabel("Marks")
 plt.title("Student Marks Analysis")
 plt.tight_layout()
 plt.savefig("student_marks.png")
 plt.show()

 # 5. Generate report
 with open("analysis_report.txt", "w", encoding="utf-8") as file:
        file.write("STUDENT DATA ANALYSIS REPORT\\n")
        file.write(f"Number of students: {len(df)}\\n")
        file.write(f"Average marks: {average:.2f}\\n")
        file.write(f"Highest marks: {highest}\\n")
        file.write(f"Lowest marks: {lowest}\\n")
        file.write(f"Top student: {top_student['Name']}\\n")

 print("\\nReport saved as analysis_report.txt")
 print("Chart saved as student_marks.png")
```

OUTPUT:-

Average Marks: 86.6

Highest Marks: 95

Lowest Marks: 75

Top Student:

Name Sita

Age 20

Marks 95